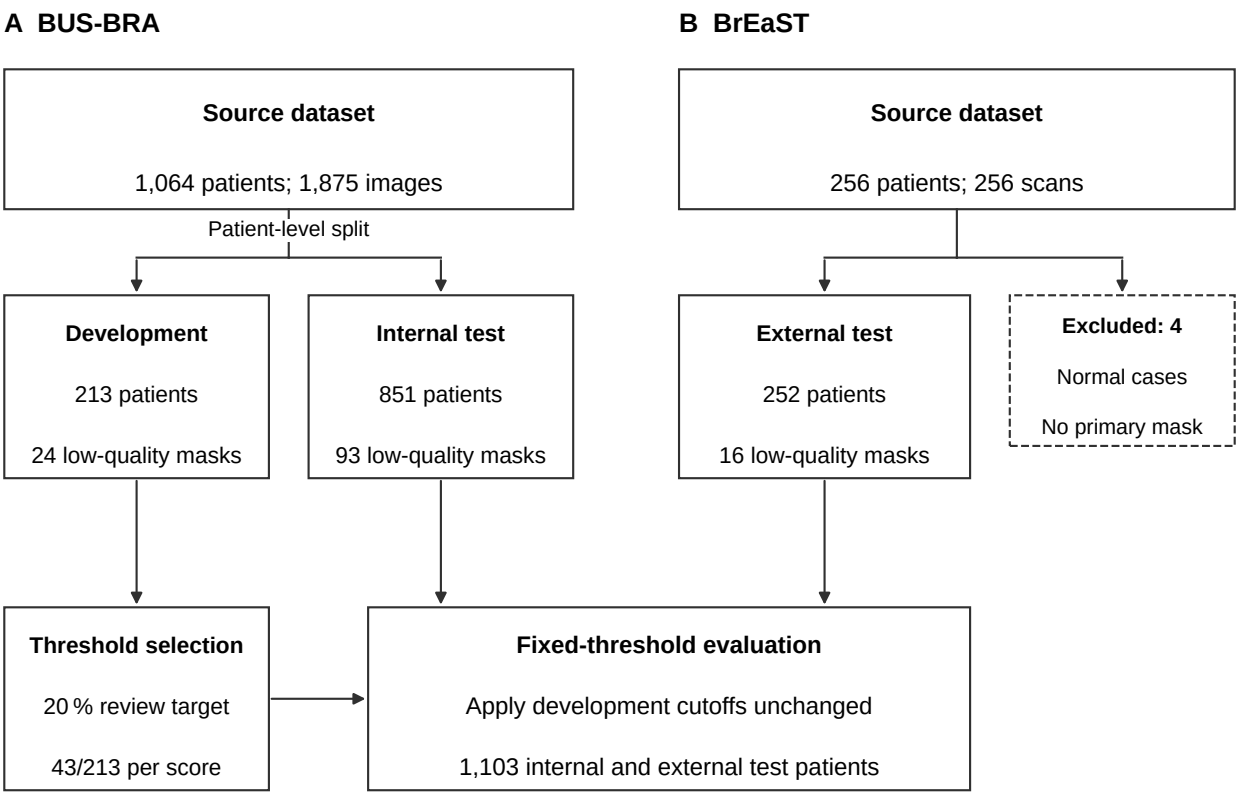
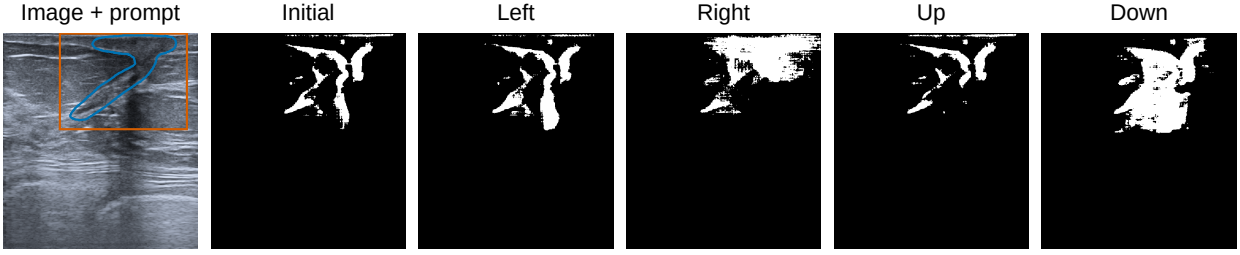


One image per patient in every analysis cohort.

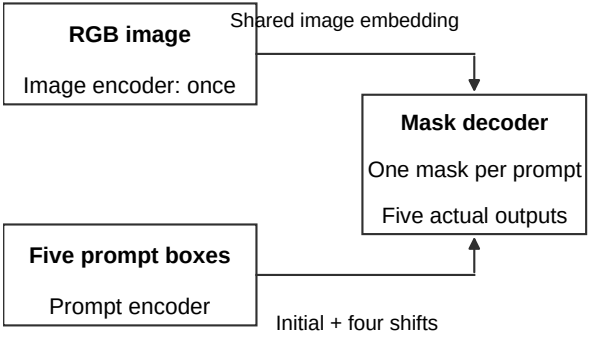
Low quality: initial-prediction Dice against the reference < 0.70.

A Image, reference-informed prompt and observed masks



BrEaST case001 | Blue: released reference | Orange: initial box | White: mask foreground

B Frozen SAM ViT-B inference



Box expansion: 10 % per side; clipped shifts: ±5 %.

Quality target: Dice(initial prediction, manual reference) < 0.70.

C Separate review scores

Method A: model confidence

Risk A = 1 − predicted IoU of initial mask

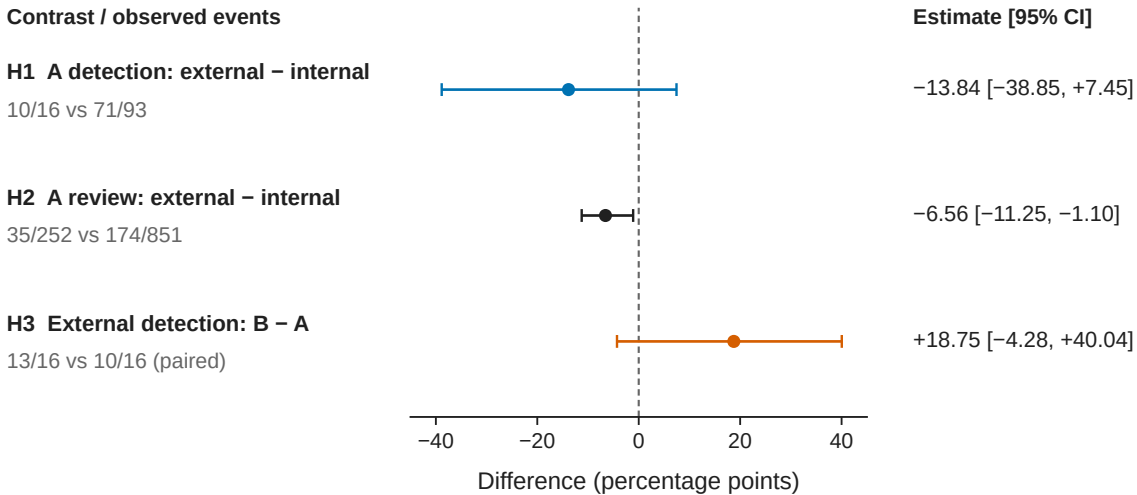
Method B: prompt disagreement

Risk B = 1 − mean Dice(initial, shifted masks)

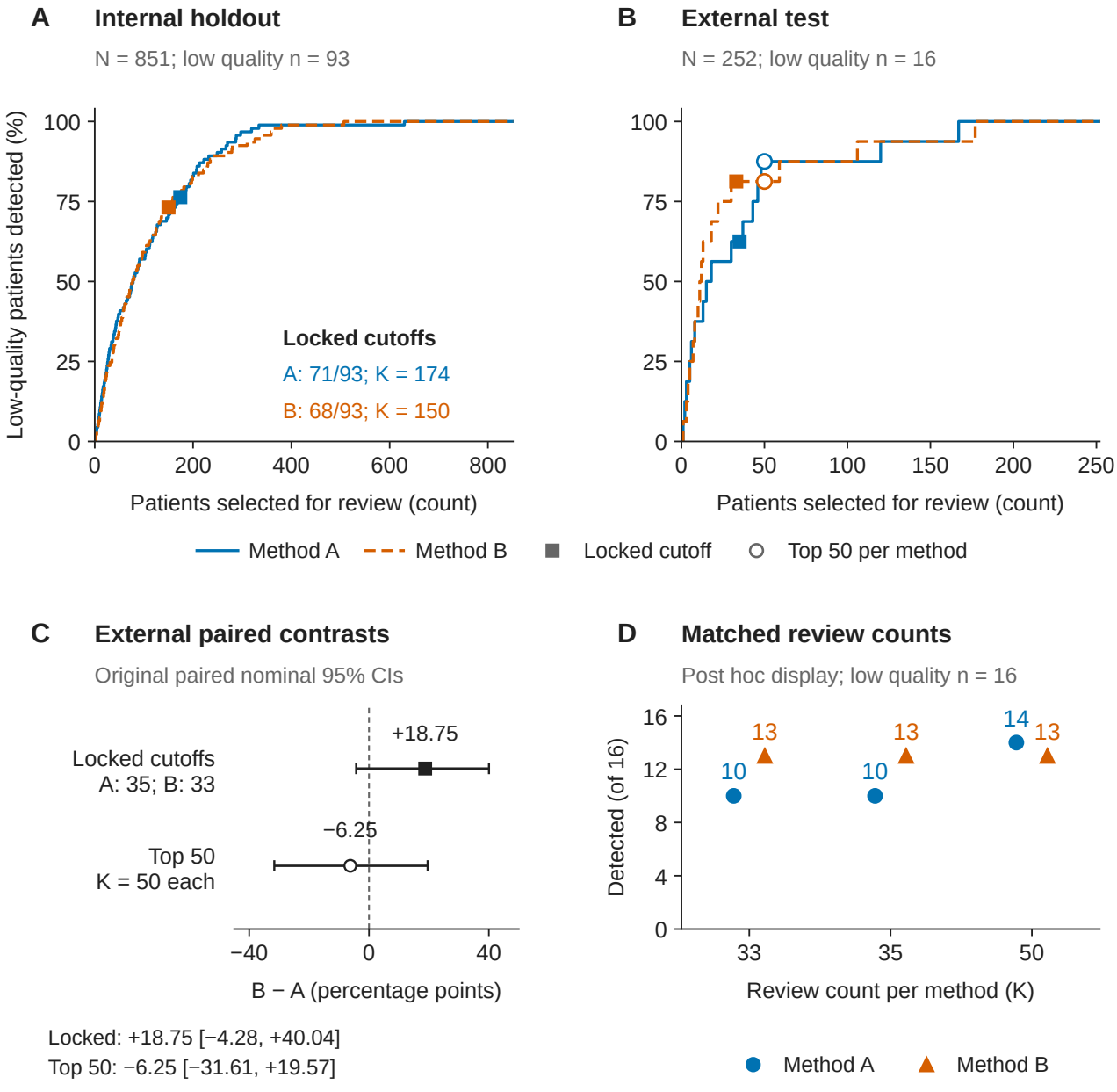
Evaluate each score at its own locked cutoff.

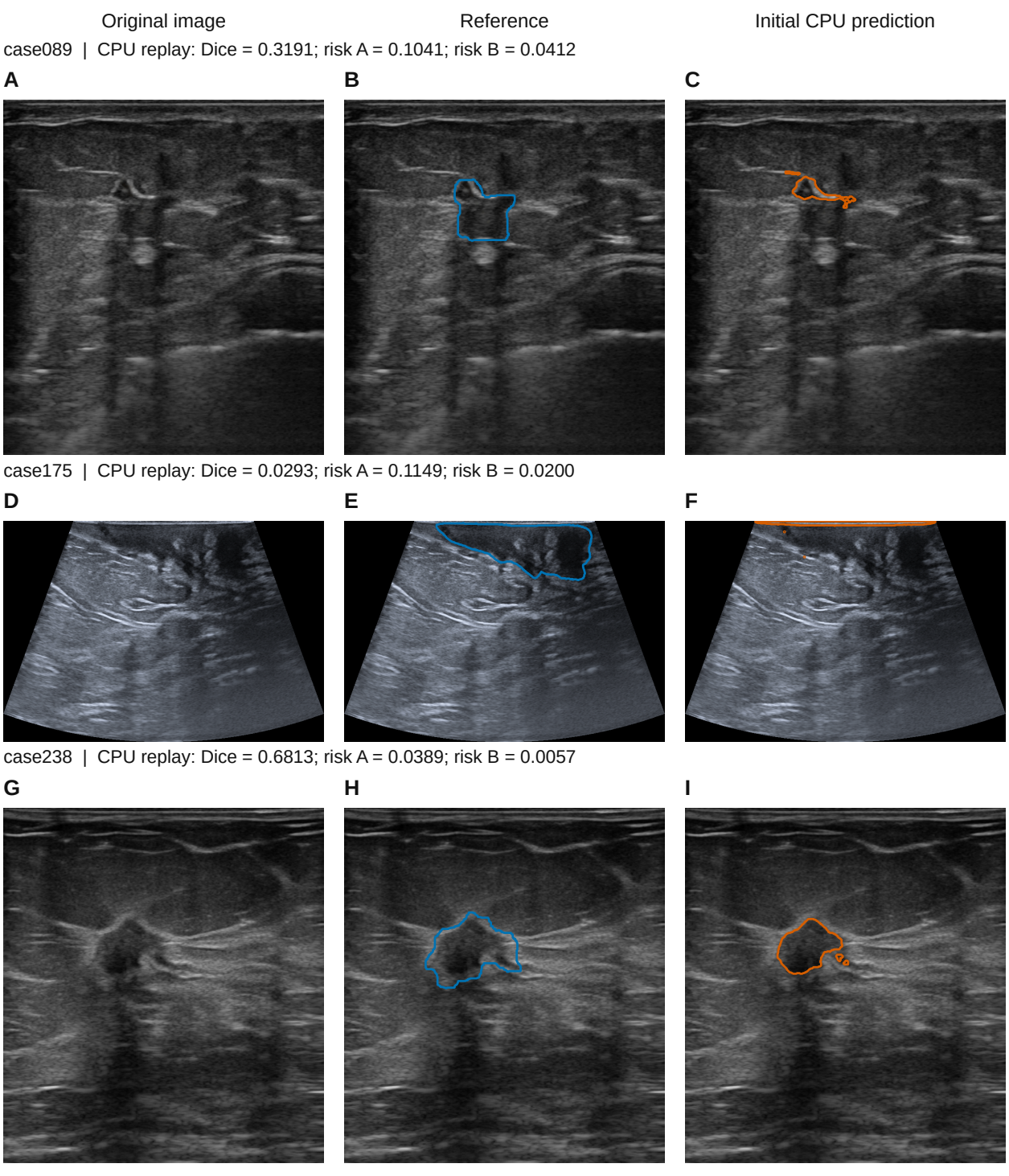
Review flag = 1 if risk ≥ cutoff.

Prespecified contrasts

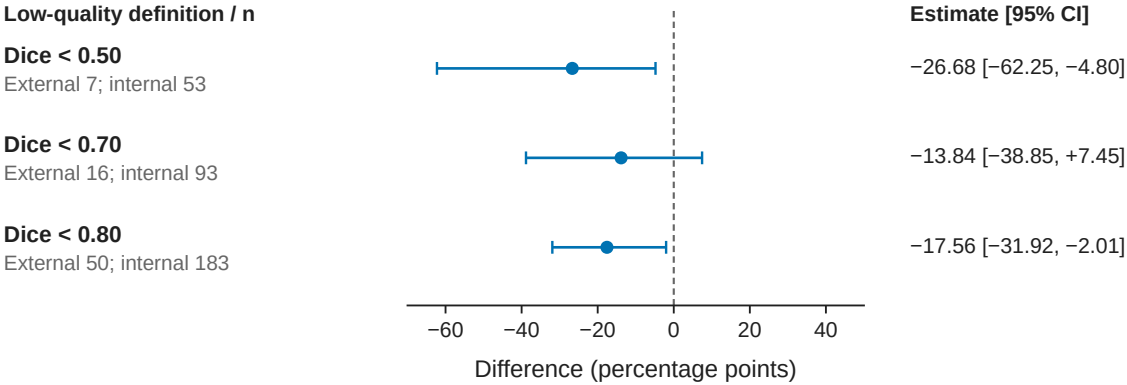


Newcombe method 10; H2 and H3 intervals are nominal.

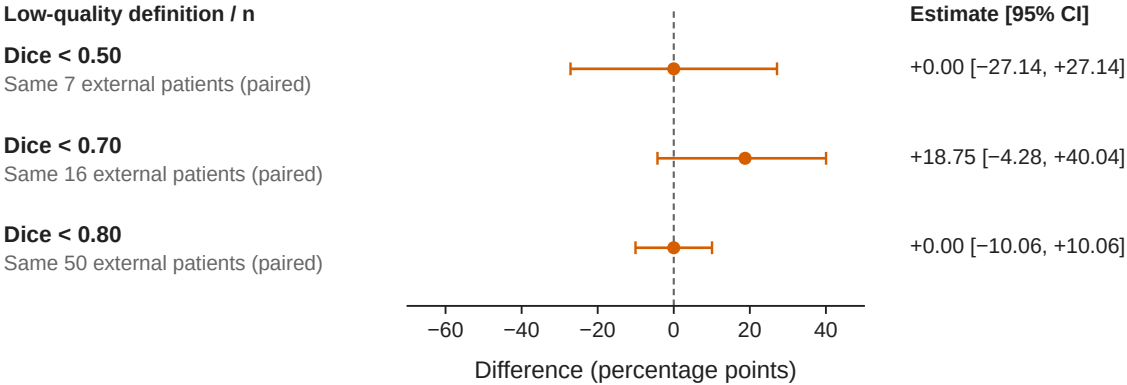

A
Internal holdout
N = 851; low quality n = 93
Low-quality patients detected (%)
100
75
50
25
0
0
200
400
600
800
Patients selected for review (count)
Locked cutoffs
A: 71/93; K = 174
B: 68/93; K = 150
B
External test
N = 252; low quality n = 16
100
75
50
25
0
0
50
100
150
200
250
Patients selected for review (count)
Method A
Method B
Locked cutoff
Top 50 per method
C
External paired contrasts
Original paired nominal 95% CIs
Locked cutoffs
A: 35; B: 33
+18.75
Top 50
K = 50 each
−6.25
−40
0
40
B − A (percentage points)
Locked: +18.75 [−4.28, +40.04]
Top 50: −6.25 [−31.61, +19.57]
D
Matched review counts
Post hoc display; low quality n = 16
Detected (of 16)
16
12
8
4
0
10
13
10
13
14
13
33
35
50
Review count per method (K)
Method A
Method B


Original image
Reference
Initial CPU prediction
case089 | CPU replay: Dice = 0.3191; risk A = 0.1041; risk B = 0.0412
A
B
C
case175 | CPU replay: Dice = 0.0293; risk A = 0.1149; risk B = 0.0200
D
E
F
case238 | CPU replay: Dice = 0.6813; risk A = 0.0389; risk B = 0.0057
G
H
I
Reference contour
CPU prediction contour

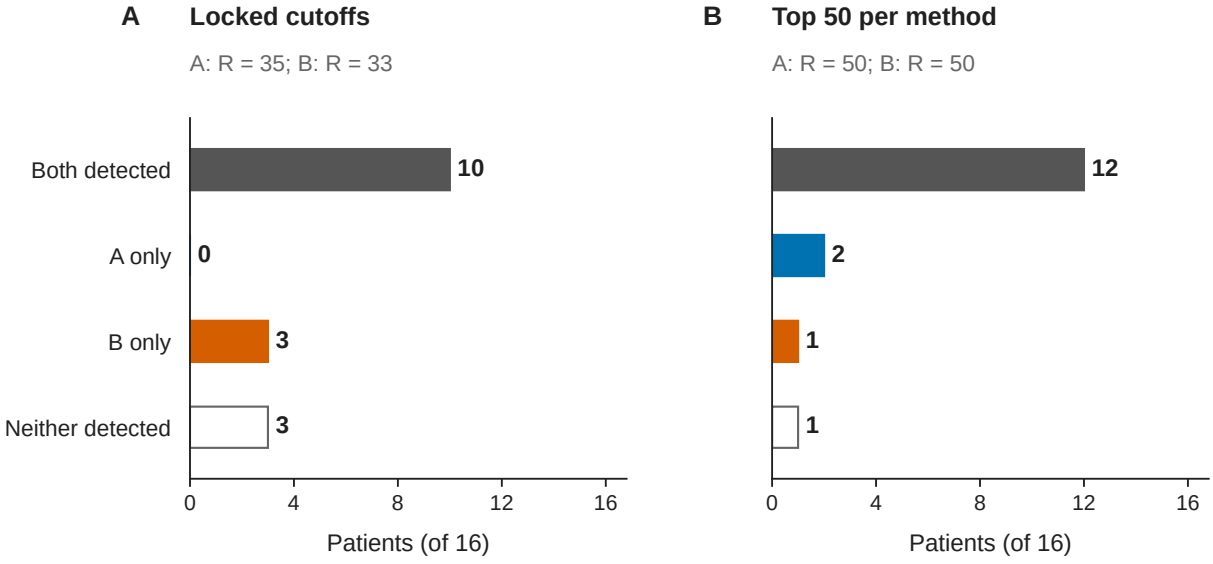
A Method A detection: external − internal



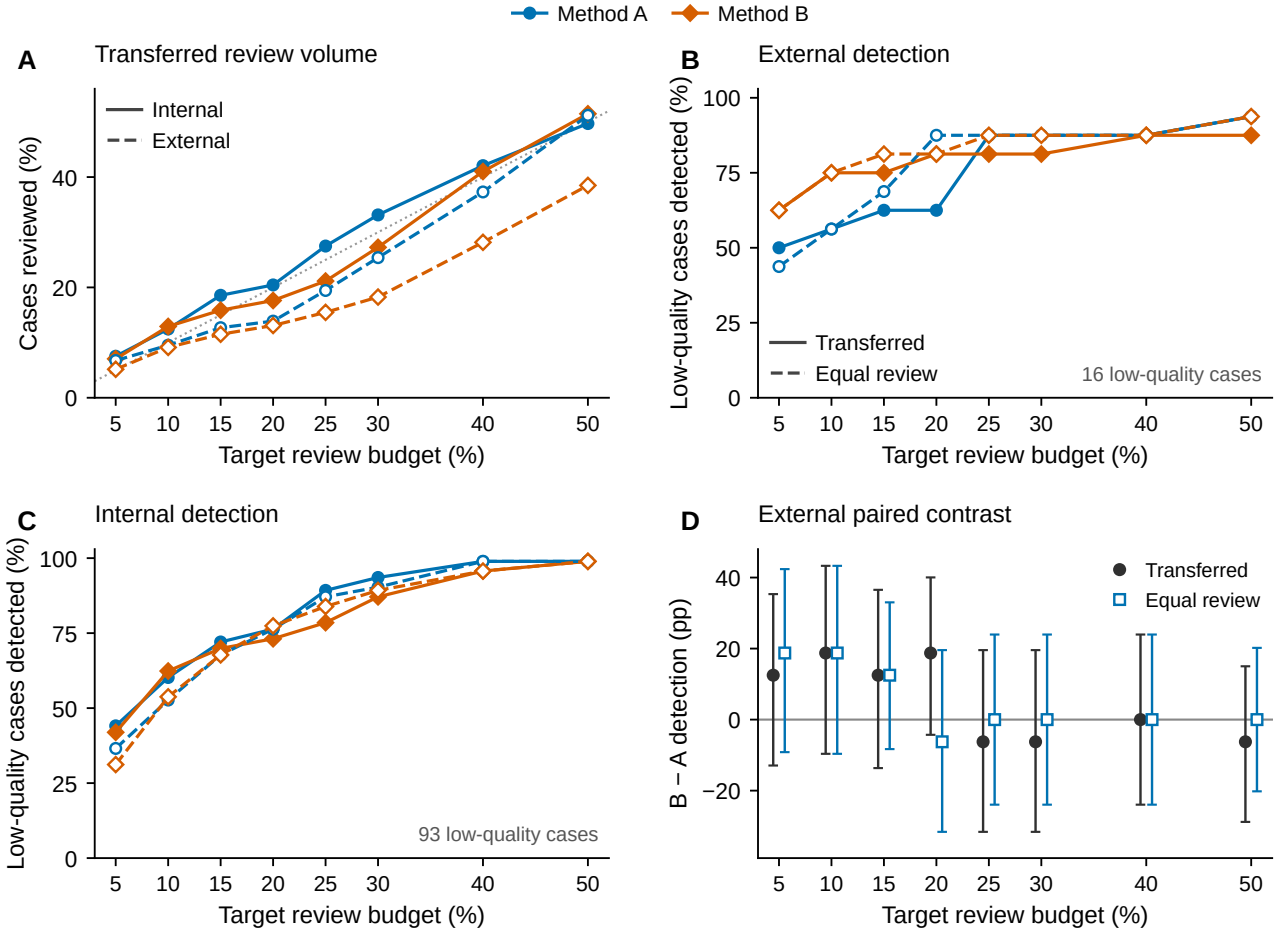
B External detection: B − A

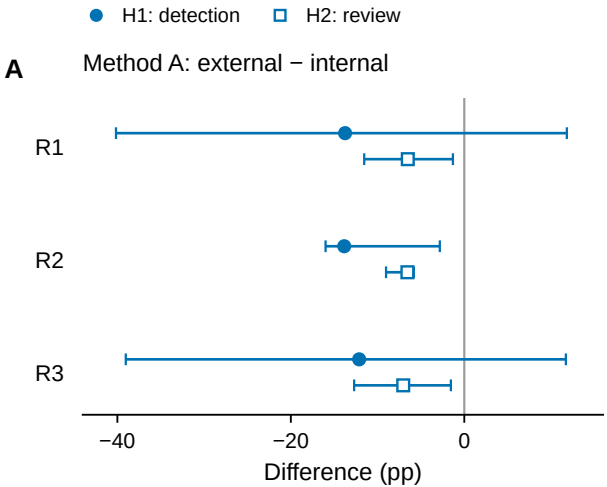


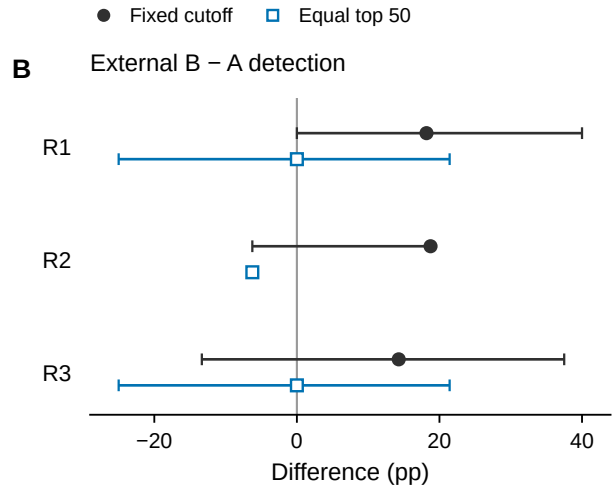
Locked risk cutoffs retained throughout; original nominal 95% Newcombe method 10 intervals.



Same 16 external low-quality patients (Dice < 0.70) in both panels.


Method A
Method B
A
Transferred review volume
Internal
External
Cases reviewed (%)
0
20
40
5
10
15
20
25
30
40
50
Target review budget (%)
B
External detection
Low-quality cases detected (%)
0
25
50
75
100
Transferred
Equal review
16 low-quality cases
Target review budget (%)
C
Internal detection
Low-quality cases detected (%)
93 low-quality cases
Target review budget (%)
D
External paired contrast
B – A detection (pp)
−20
0
20
40
Transferred
Equal review
Target review budget (%)

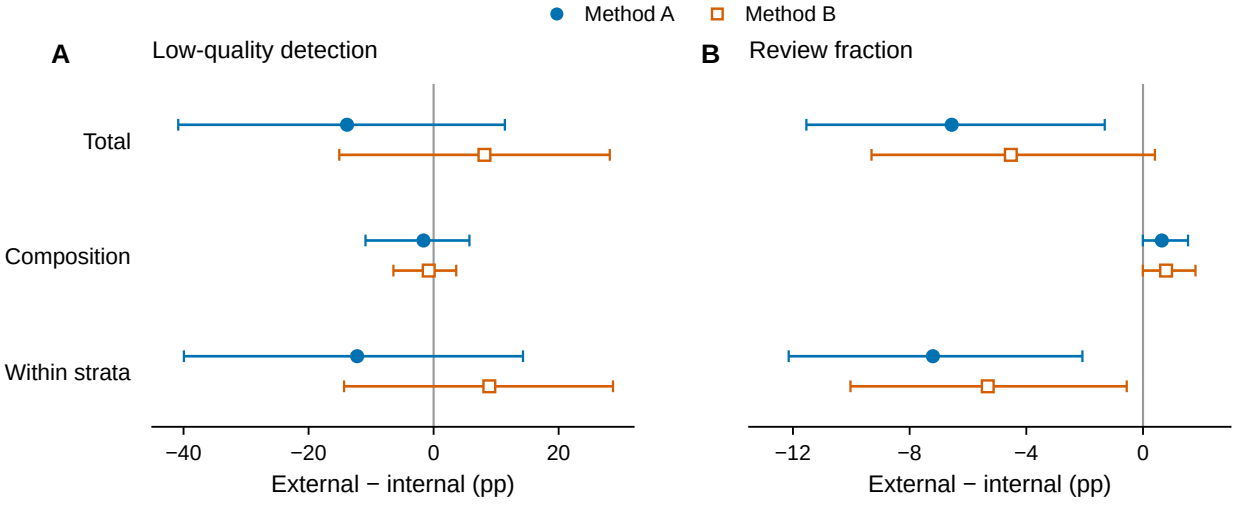


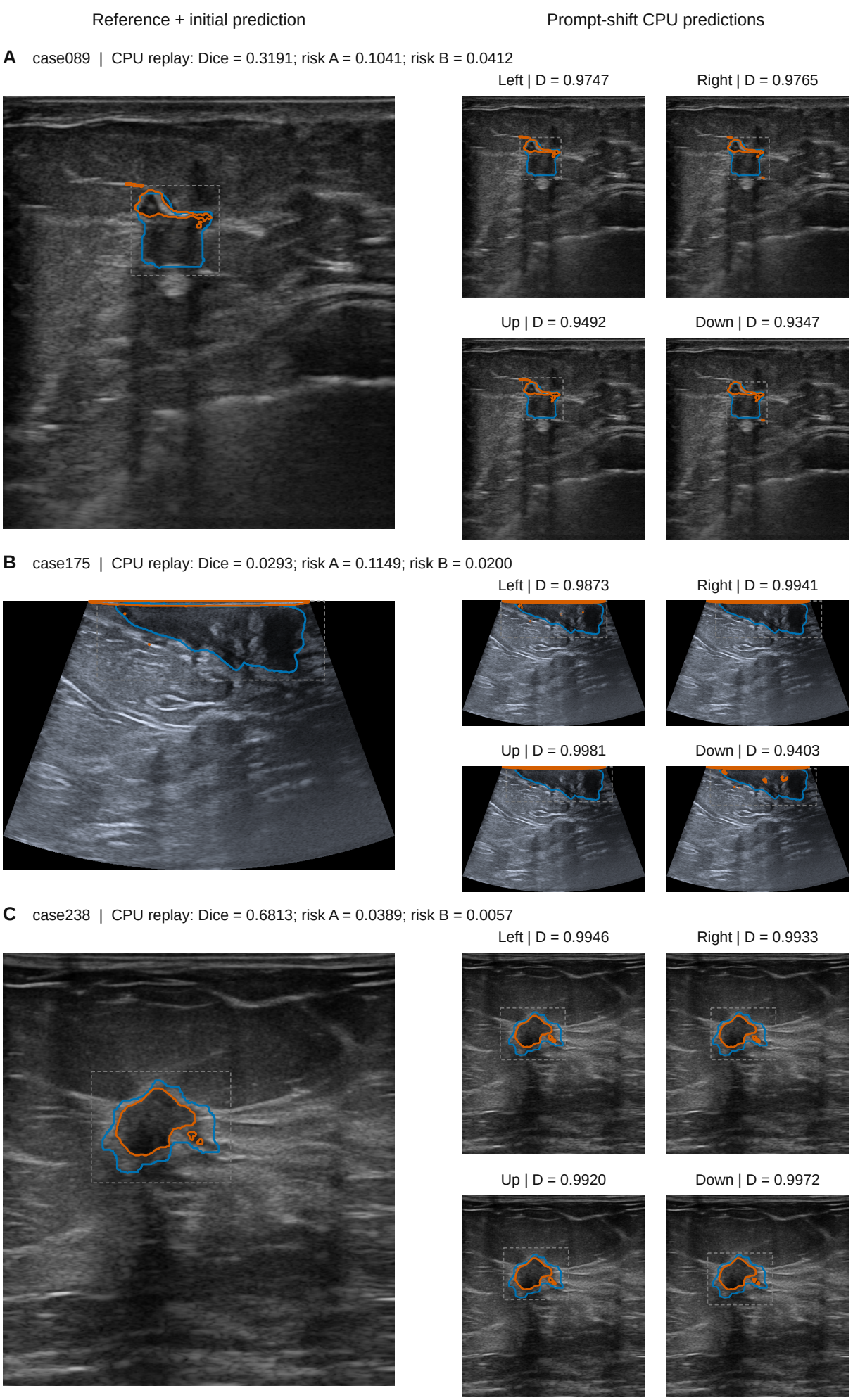


C

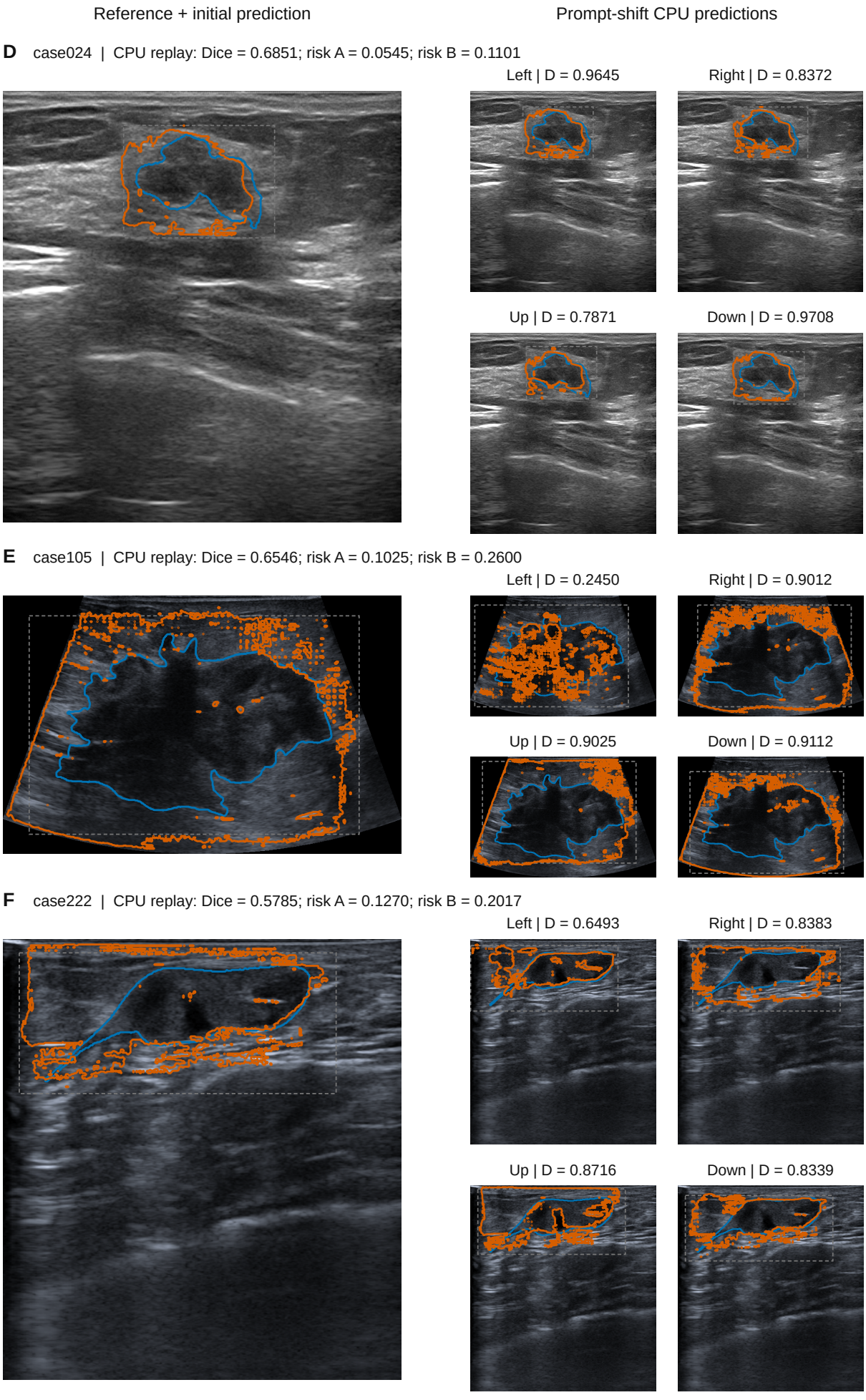
Joint fixed / top-50 sign frequencies

	Original strict	Opposite strict	Any zero	Same nonzero
R1	39.61 %	0.00 %	25.19 %	35.20 %
R2	90.40 %	0.00 %	4.35 %	5.25 %
R3	30.94 %	0.90 %	28.08 %	40.08 %


Method A
Method B
A
Low-quality detection
Total
Composition
Within strata
−40
−20
0
20
External − internal (pp)
B
Review fraction
−12
−8
−4
0
External − internal (pp)



D: Dice vs initial CPU mask. All images retain the full original field of view.



D: Dice vs initial CPU mask. All images retain the full original field of view.